

Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

[Project Proposal]

Project Title: AI-Powered Resume Analyzer and Career Copilot

Course Title: Mobile Computing Lab

Course Code: CSE4204

Submitted to:

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TEAM INFORMATION SHEET

Field	Details
Official Team Name	CSE4204-8A-T05
Section	8A
Project Title	AI-Powered Resume Analyzer and Career Copilot
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GitHub Repository Link	https://github.com/mahamud894/AI_Resume_Analyzer

AI-Powered Resume Analyzer and Career Copilot

1. PROBLEM STATEMENT:

- **The Existing Problem:** Modern recruitment processes rely heavily on Applicant Tracking Systems (ATS). Capable candidates and fresh graduates are frequently filtered out and rejected automatically due to poor resume formatting, weak action verbs, or a lack of targeted keywords from job descriptions.
- **Why the Problem is Important:** Job hunting is highly competitive. Without proper insights, applicants face continuous, repetitive rejections without knowing what to fix, while manually writing personalized cover letters for each individual job application remains highly time-consuming.
- **Why Existing Solutions are Insufficient:** Most available online ATS platforms are highly expensive, require complex setups, or offer rigid keyword-matching without contextual understanding or actionable career steps.
- **Why AI is Relevant:** AI is essential to contextually parse resumes, evaluate semantic skill sets against job descriptions, and provide resume analysis, ATS scoring, smart recommendations, and human-like personalized rewrite suggestions in a single platform.

2. OBJECTIVES:

- **Establish a Client-Side Serverless Environment:** Setup a modern React.js frontend integrated with the Puter.js SDK to bypass traditional Express.js/Node.js backend configurations and eliminate API key exposure.
- **Implement Secure Serverless Authentication:** Build a complete user onboarding and session management workflow utilizing Puter's native secure login/logout methods directly from the UI.
- **Develop a Document Vault & Versioning System:** Implement an interactive dashboard containing a drag-and-drop file uploader (via React Dropzone) to store, retrieve, and delete resume versions using Puter's NoSQL storage.
- **Integrate Asynchronous PDF-to-Image Parsing:** Convert uploaded multi-page PDF resumes into background images seamlessly to generate clean visual previews on the user dashboard.
- **Deploy Context-Aware ATS Evaluation:** Utilize advanced LLMs (Claude 3.5 Sonnet / GPT-4o) through Puter AI to cross-reference extracted resume data against job descriptions, calculating an instantaneous ATS match score across Tone, Structure, Content, and Skills.

- **Automate Career Asset Generation:** Integrate generative AI modules to instantly produce highly tailored professional cover letters and contextual mock interview questionnaires derived from detected resume flaws.
- **Achieve Zero-Cost Cloud Deployment:** Deploy the completed application on Puter's unified hosting infrastructure, proving the feasibility of a high-performance "User-Pays" production app with zero developer server overhead.

3. PROPOSED FEATURES:

- **Authentication System:** Secure user sign-in and account session management via the Puter API framework.
- **Interactive Analytics Dashboard:** A responsive visual control panel tracking past resume revisions, upload history, and optimization metrics featuring visual score gauges.
- **Secure Document Vault:** Serverless cloud file hosting to safely store, track, and version multiple resume documents (PDF/DOCX).
- **Semantic Resume Parser & Score Meter:** AI-based, context-aware scanning of uploaded PDF data to evaluate layout, vocabulary alignment, and job description matching.
- **AI Cover Letter Generator (Advanced Feature):** Automatically writes a tailored, highly optimized cover letter by analyzing the uploaded resume against targeted job descriptions.
- **Document Export Module:** PDF export functionality to seamlessly download generated assessment reports and documents.

4. TECHNOLOGY STACK:

Component	Technology	Functional Description
Frontend	React.js with React Router V7 + Tailwind CSS	For building a highly responsive, clean, and modern user interface.
Backend & Cloud	Puter.js (Serverless Architecture)	Handles user auth, application logic, and cloud functions natively from the client side without standalone server infrastructure.

Component	Technology	Functional Description
Database	Puter NoSQL & KV Storage	For lightning-fast user history and document metadata tracking with zero-config setup.
File Storage	Puter Built-in File Hosting API	For saving, hosting, and managing resume files safely.
Programming Language	TypeScript	Use to ensure clean,type-safe code through the project.
State Management	Zustand	Use for simple and efficient global state management
AI Integration	Puter.js AI Module GPT-4o / Claude 3.5 via Puter AI	Directly executing advanced Large Language Models contextually without exposing structural API keys.
PDF Processing	pdfjs / react-pdf	Context-aware scanning of file data and rendering downloadable reports.
Development Tool	Vite	Used to spin up and manage the React project
Development Environment	Webstorm	Project developed in Webstorm
Deployment	Vercel / Puter Hosting	Cloud deployment platform ensures absolute zero server infrastructure overhead.

5. SIMILAR EXISTING SYSTEMS / RESEARCH:

1. Jobscan / Resume Worded: Commercial platforms providing resume auditing against job listings.

Limitation: Highly premium, lacks instant serverless responsiveness, and does not provide integrated real-time cover letter formatting or tailored interview preparations on a single dashboard.

2. Reactive Resume: An open-source resume builder.

Limitation: It helps build resumes manually but completely lacks intelligent ATS tracking, multi-model AI checking, and automated pre-interview assistant loops.

6. EXPECTED OUTCOME:

- **User Gains:** Job seekers receive an all-in-one ecosystem that not only critiques their resumes but also formats supporting application materials (Cover letters) and trains them for prospective interviews.
- **Workflow Improvement:** It condenses hours of manual research, cross-referencing, and preparation into an intuitive, seconds-long automated workflow, saving time and increasing direct interview opportunities.
- **Benefits of AI Integration:** Moving beyond simple text-matching allows the platform to understand deep concepts, rewrite technical bullets gracefully, and protect operational privacy while operating at an infrastructure cost of absolute zero.

7. PROJECT TIMELINE:

Week	Task	Phase Deliverables
Week 1-2	Requirement Analysis & UI Design	Requirement analysis, architectural design, and UI wireframing.
Week 3-4	Frontend Development	Building responsive React components and layout systems with Tailwind CSS.
Week 5-6	Authentication & Resume Upload	Integrating Puter secure authentication and serverless file upload support.
Week 7-8	ATS Analyzer Integration	Embedding semantic pdf-parse tooling and deep-level ATS score analyzer tracking.

Week	Task	Phase Deliverables
Week 9-10	AI Cover Letter Generator	Developing the AI automated cover letter generator matching job description data.
Week 11-12	Mock Interview Generator	Developing the contextual mock interview generator and dashboard metrics.
Week 13-14	Testing & Deployment	Complete system testing, debugging, optimization, and cloud deployment.
Week 15	Final Presentation	Compiling final reports, system verification, and final project presentation.

8. CONCLUSION:

The AI-Powered Resume Analyzer & Career Copilot is a practical and innovative semester project that efficiently combines resume analysis, ATS scoring, AI-generated cover letters, and interview preparation into one intelligent platform. The project directly solves real-world career challenges and showcases meaningful AI integration while remaining perfectly feasible for semester-long development.